# Claudespeak: Characterizing the Stylistic Fingerprint of a Frontier Language Model

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## Abstract

Large language models are widely believed to have recognizable writing styles, but most popular claims (e.g. that models overuse the word *delve*) are derived from *ChatGPT* and applied indiscriminately to all models. We ask what, specifically, characterizes the prose of one frontier model—Anthropic's Claude Opus 4.8—relative to human writing and to seven other models, holding topic constant via parallel corpora. Combining interpretable stylometry, log-odds $n$-gram selection, a length-controlled regression, an interpretable classifier, and suffix-automaton ($n$-gram novelty) mining, we find a stable fingerprint that is *structural, rhythmic, and interactional* rather than a matter of stereotyped vocabulary. Claude is identifiable from prose alone (prose-only classifier AUC $0.95$ vs. a human/GPT-4o/GPT-3.5 panel; $0.87$ vs. six modern models), driven by markedly higher sentence-length variability, denser content-word usage, a strong em-dash habit, and—most diagnostically—a verbatim offer-to-continue closing move ("would you like me to go deeper into…") that, in our corpus, appears dozens of times in Claude and *zero* times across humans and all other models. We further show that the stereotyped "delve" vocabulary is a GPT (and other-model) trait that Claude *avoids*: across seven models in an essay genre, Claude is the only model that never writes *delve* and uses *crucial* least. We release the fully provenanced corpus and pipeline. Sections on the effect of reasoning effort, Claude versions, and multi-turn dynamics are forthcoming.


## 1 Introduction

The claim that text "sounds like AI" is now commonplace, and a folk vocabulary of tells—*delve*, *tapestry*, *crucial*, *intricate*—circulates widely. Two problems undercut these claims as characterizations of any *particular* model. First, the evidence base is overwhelmingly about ChatGPT (Kobak et al., 2025), yet the conclusions are applied to "LLMs" generically. Second, most analyses do not control topic: a classifier that distinguishes model from human text may be learning the prompt distribution, not the writing style.

We study a single frontier model, Anthropic's Claude Opus 4.8, and ask a deliberately narrow question: *holding the prompt fixed, what makes Claude's prose identifiable relative to humans and to other models?* We build two topic-matched parallel corpora in which the same prompts are answered by Claude, by humans, and by up to six other models, and we attack the question with five complementary methods. Our contributions:

1. A parallel-corpus protocol and a fully provenanced dataset (every generation stored with model, decoding configuration, token usage, and timestamp) that supports reuse without regeneration.
2. Evidence that Claude's fingerprint is *structural and rhythmic* (sentence-length burstiness, content-word density, em-dashes) and survives both markdown stripping and length control.
3. Identification of a verbatim *interactional* signature—an offer-to-continue closer—via log-odds and suffix-automaton mining, present in Claude and absent from all other sources in our corpus.
4. A reconciliation of the "delve" folklore: the stereotyped vocabulary is a GPT/other-model trait; Claude is the cleanest frontier model on it.
5. A *coverage taxonomy* (intent $\times$ interaction structure $\times$ stance) and a reproducible audit showing that standard corpora occupy a narrow slice (single-turn, cooperative, informational); we argue that extending model char-

acterization into the under-observed cells—multi-turn corrective, contested-stance, sensitive, emotional—is itself a contribution, and we release the audit tooling.

## 2 Related Work

**Detecting machine-generated text.** A large literature aims to decide *whether* a text is machine-generated. Supervised detectors fine-tune classifiers such as RoBERTa on model outputs (Solaiman et al., 2019), while zero-shot methods exploit statistical signatures of the generating distribution: probability curvature (Mitchell et al., 2023) or the cross-perplexity of a model pair (Hans et al., 2024). Dugan et al. (2024) show such detectors are brittle across domains and attacks, and watermarking (Kirchenbauer et al., 2023) sidesteps detection by injecting a signal at generation time. We do not detect machine text; we instead *characterize* the style of one frontier model, assuming provenance is known.

**Attributing text to a model.** Closer to our aim is work on *which* model wrote a text. Surveys frame this as authorship attribution in the LLM era (Huang et al., 2024), and discriminative classifiers can separate model families. We complement attribution: rather than optimizing a label, we use an interpretable, length-controlled regression and log-odds $n$-gram selection (Monroe et al., 2008) to surface *what* distinguishes Claude, and we audit which stylistic claims hold up under corpus coverage.

**Excess vocabulary and "AI-ese."** Kobak et al. (2025) quantify LLM-marker words (e.g. "delve") as excess vocabulary in the biomedical literature; popular accounts trace the habit to RLHF annotation pipelines (Willison, 2024). Such word-lists, however, are largely a ChatGPT/GPT signature. We treat the "delve" folklore as a hypothesis to test rather than a given: our markers are discovered data-drivenly from parallel corpora, and we show this trait is *not* characteristic of Claude.

**Stylometry and $n$-gram novelty.** Our features build on classical stylometry, notably function-word frequencies and Burrows's Delta (Burrows, 2002). To probe memorization versus genuine novelty, we adopt suffix-automaton $n$-gram search (Merrill et al., 2024) and unbounded $n$-gram indexing (Liu et al., 2024), applying novelty mining to a single model's outputs comparatively.

| Track | Prompts | Sources (same prompt each) |
|---|---|---|
| HC3 | 200 | human, ChatGPT (GPT-3.5 era), GPT-4o†, **Claude**† |
| AlpacaEval | 200 | gpt-4-turbo, gpt-4o, gemini-pro, deepseek-67b, Qwen2-72B, Llama-3-70B, **Claude**† |

Table 1: The two parallel corpus tracks. † = self-generated; all other cells reused. HC3 carries a human anchor; AlpacaEval carries the modern multi-model contrast (no human, by design). HC3 spans five source domains (reddit-ELI5 and open-QA are merged into "general" for reporting).

**Sycophancy and persona.** Frontier models exhibit measurable behavioral signatures: sycophancy and agreement under pressure (Perez et al., 2022; Sharma et al., 2023), and self-interaction attractors such as the "spiritual bliss" state reported for Claude (Anthropic, 2025). These motivate our future-work agenda on multi-turn corrective behavior, where stylistic fingerprints may interact with concession dynamics.

**Data.** We draw on parallel human/LLM corpora including HC3 (Guo et al., 2023) and AlpacaEval (Li et al., 2023), repurposing them for comparative stylometric characterization rather than detection or preference ranking.

## 3 Corpus

We construct two parallel tracks (Table 1). In each, a fixed set of prompts is answered by multiple sources, so that any feature separating Claude reflects *how* it writes rather than *what* it was asked. Claude (and GPT-4o on the HC3 track) are generated by us with adaptive thinking at `effort=high`; all other cells are reused from the source datasets. Every record stores full provenance—model and version, decoding configuration, token usage, timestamp, and code commit—plus the verbatim API response. The seven distinct non-Claude models are GPT-3.5-era ChatGPT, GPT-4o, GPT-4-turbo, Gemini-pro, DeepSeek-67B, Qwen2-72B, and Llama-3-70B; with the human anchor this gives eight comparison sources (six of which—the "modern models"—appear on the AlpacaEval track).

### 3.1 Corpus statistics

Table 2 summarizes volume and length. Responses are model-typical in length (Claude median 230–267 words); the human HC3 answers

| Track | Prompts | Prompt wds (md) | Claude wds (md) | Total words |
|---|---|---|---|---|
| HC3 | 200 | 10 | 267 | 156k |
| AlpacaEval | 200 | 18 | 230 | 354k |

Table 2: Corpus volume and length (median word counts). Totals sum all sources in the track. Full per-source distributions: Appendix B.

are much shorter and more variable (median 77, mean 120), which we account for via length controls (§5.3). Prompts are short (median 10–18 words) and roughly half are phrased as questions. Per-domain volume and full distributions are in Appendix B. Two corpus-level register signals already foreshadow the results: Claude's responses are markedly more *interrogative* at the sentence level (17% of sentences on HC3, 10% on AlpacaEval, vs. $\leq$4% for every other source—the offer-closer, §5.4), and least readable (Flesch $\approx$40 vs. 46–52), consistent with higher content-word density.

### 3.2 Coverage and its limits

Because style is context-dependent, we audit *what behavior the prompts elicit*, labeling every prompt by intent, speech act, register, tone, and topic, plus interaction flags (`src/analyze_coverage.py`; Table 3). Labels are produced by Claude Opus 4.8 itself (the model under study), a circularity we flag in Limitations.

TODO (human validation pending): hand-annotate a $\sim$50-prompt sample, report inter-annotator agreement with the LLM labels, and add a small validation table. Quick human pass to be added.

The corpus occupies a narrow region of the behavior space along several axes:

- **Mood and speech act.** We measure both the *syntactic mood* (rule-based, from punctuation and sentence-initial cues) and the *pragmatic speech act* (LLM-labeled), which differ because many syntactically interrogative prompts ("Can you write. . . ?") are pragmatic directives. Syntactically, prompts are 74%/58% (HC3/AlpacaEval) interrogative, 2%/20% imperative, $\sim$22% declarative; pragmatically they are questions or directives with $\leq$1 *assertion* per track (Appendix B). Either way, the corpus almost never presents Claude with an assertion to agree with, qualify, or contest.

| Intent (primary) | HC3 | AlpacaEval |
|---|---|---|
| explanation | 94 | 33 |
| factual QA | 53 | 29 |
| advice/recommendation | 49 | 31 |
| instruction task | 3 | 26 |
| creative writing | 0 | 24 |
| math/reasoning | 0 | 14 |
| code | 0 | 11 |
| opinion/subjective | 0 | 11 |
| other (classif., edit, roleplay) | 1 | 20 |
| *flag:* multi-turn implied | 8 | 4 |
| *flag:* subjective | 14 | 39 |
| *flag:* sensitive | 61 | 16 |

Table 3: Prompt intent distribution and interaction flags (HC3 $n$=200; AlpacaEval $n$=199, one prompt unlabeled). Single frozen audit run (`coverage_labels.csv`). The corpus is single-turn, cooperative, and information-oriented.

- **Interaction.** Only $\sim$2% of prompts imply multi-turn interaction, and essentially none are adversarial or corrective.
- **Intent.** Code, creative, opinion, and emotional intents are rare or absent, especially in HC3 (Table 3).
- **Stance / register / tone.** Prompts are predominantly neutral in tone and information-seeking; subjective or contested-stance prompts are a small minority.

Consequently the fingerprint we report should be read as Claude's *single-turn informational* voice. Behaviors gated on other cells—e.g. the concession move "you're right to push back on that," which requires a *declarative* challenge across turns—are out of scope here and motivate the corpus expansion (more prompt variety across topic, register, tone, speech act, and interaction structure) that we describe as future work (§5.8, Appendix B).

## 4 Methods

We compute interpretable features per response (lexical: type-token ratio, hapax rate, function-word rate; punctuation: em-dash, colon, question, exclamation, emoji; syntax: mean sentence length and *burstiness*, the ratio of the standard deviation to the mean of words-per-sentence; markdown/structure; and rhetoric). Rates are per 100 or 1000 words so that text length does not mechanically dominate. We compute two dozen such features per response. We summarize source differences with Cohen's $d$ (Claude vs. a compari-

| Prose feature | $\lvert d \rvert$ | Claude / others |
|---|---|---|
| Sentence burstiness | 1.69 | 0.86 / 0.36–0.55 |
| Function-word rate ↓ | 1.47 | 0.33 / 0.37–0.44 |
| Em-dash rate | 0.87 | 0.88 / ≤0.40 |
| Colon rate | 0.84 | 1.41 / 0.24–1.69 |
| Question rate | 0.82 | 0.61 / 0.01–0.28 |

Table 4: HC3 track, Claude vs. pooled others, markdown stripped. Mean |Cohen's $d$| and group means (Claude / range over the other sources). ↓ = Claude lower.

**Prompt (ELI5):** *Why is skin itchy?*
**Human:** "As far as I know, scratching is an evolved reaction to help stop insects and other parasites from sucking our blood. . . "
**Claude Opus 4.8:** ". . . scratching sends a *new* signal to your brain that covers up the itchy feeling. But scratching too much can hurt your skin, so it's better to be gentle! **Would you like me to explain anything else?**" [closes with an emoji]

Figure 1: Same prompt, two sources (excerpts). Claude's answer is longer, more structured, and ends with the offer-to-continue closer; the human answer is terse and declarative.

son group); all reported effects are significant at $p < 0.001$ (Welch's $t$, $n$=200 Claude vs. ≥200 comparison). The classifier is $\ell_2$-regularized logistic regression ($C$=1.0, balanced class weights). To separate *voice* from *formatting*, we additionally compute all prose features after stripping markdown markup. We mine characteristic $n$-grams with the Monroe et al. (2008) log-odds estimator; quantify length-independence with OLS (§5.3); measure separability with $\ell_2$-regularized logistic regression under 5-fold cross-validation; and mine arbitrary-length spans and $n$-gram novelty with a suffix automaton (Merrill et al., 2024).

# 5 Results

## 5.1 A large signal, dominated at first by formatting

The strongest raw separators between Claude and the other HC3 sources are markdown-structural: markdown headers ($d$=2.9), bold ($d$=2.3), and bullets ($d$=1.8), which the human and old-ChatGPT cells use essentially never. This is real but is a *formatting* habit; the decisive question is whether anything survives removing it.

## 5.2 The voice survives markdown stripping

With all markup stripped, Claude still separates strongly on prose (Table 4): it interleaves short and long sentences far more than any other source (burstiness), writes denser content-word prose (lower function-word rate), and retains a marked em-dash habit. These are the load-bearing stylistic findings.

## 5.3 The signature is not a length artifact

On the HC3 track, Claude writes longer answers (mean 253 words vs. 120 for humans, 228 for GPT-4o), so we test whether the prose effects merely track length. Within every word-count quartile the effects persist, and a length-controlled regression (HC3) (feature ∼ is_claude + $z$(n_words)) leaves the Claude coefficient large and highly significant: burstiness $t$=20.6, function-word rate $t$=−14.4, em-dash $t$=12.6, question rate $t$=4.6. Type-token ratio is length-sensitive and we treat lexical-diversity claims as length-conditioned. The effects also hold *within every domain* (Appendix C): across all nine domains in both tracks, burstiness has $d > 0$, function-word rate $d < 0$ (Claude denser), and em-dash and question rate $d > 0$, so the fingerprint is not a topic artifact.

## 5.4 The interactional signature: an offer-to-continue closer

Data-driven $n$-gram mining (Monroe et al., 2008) surfaces a clear interactional fingerprint rather than a content-word list. The most Claude-like trigrams (HC3) are *would you like*, *like me to*, *me to explain*, *in more detail*, *to go deeper*; the anti-signature—terms Claude *underuses*—is led by the formal frame *it is important to*. At the sentence level this closer registers as a striking jump in interrogatives: **17%/10% of Claude's response sentences are questions (HC3/AlpacaEval), versus ≤4% for every other source** (Appendix B). Suffix-automaton mining (§5.7) confirms the pattern at arbitrary length. Figure 1 shows the fingerprint in a single matched pair.

## 5.5 The "delve" vocabulary is a GPT trait Claude avoids

On the AlpacaEval track—an essay/instruction genre that *does* elicit the stereotyped words—Claude is the cleanest of seven models (Table 5). It is the only model that never writes *delve* or *delves*, and it uses *crucial* least of all seven (0.039 per 1k vs. DeepSeek 0.054, Gemini 0.090, Llama 0.185, GPT-4o 0.340, Qwen 0.346, GPT-4-turbo 0.422). Claude's own lexical lean is conversational ("ac-

| per-1k | Claude | g4t | g4o | qwen | llama |
|---|---|---|---|---|---|
| delve | **0** | .018 | .018 | 0 | .017 |
| delves | **0** | .037 | .089 | .019 | .034 |
| crucial | **.039** | .422 | .340 | .346 | .185 |
| intricate | **0** | 0 | 0 | .019 | .050 |
| pivotal | **0** | .073 | .018 | .038 | 0 |

Table 5: Frequency (per 1000 words) of stereotyped "excess" words on the AlpacaEval essay track. g4t = gpt-4-turbo, g4o = gpt-4o. Claude is at or near the minimum for every term. (gemini/deepseek omitted for space; same pattern.)

| Track | with formatting | prose only |
|---|---|---|
| HC3 | 0.982 | 0.946 |
| AlpacaEval | 0.901 | 0.867 |

Table 6: Claude-vs-rest ROC-AUC (logistic regression, 5-fold CV). HC3 contrasts Claude with {human, GPT-4o, GPT-3.5-era ChatGPT}; AlpacaEval with the six modern models.



Figure 2: **[Placeholder image—replace with real plot.]** $n$-gram novelty by source: the fraction of each source's $n$-grams ($n$=1 . . . 10) that are absent from the pooled other sources (AlpacaEval track). Claude's curve sits above all six other models at every $n$ (bigram novelty 0.519 vs. 0.27–0.36), i.e. its phrasing overlaps least with the pool. Data: `reports/ngram_dawg.md`.

Replace the grey placeholder with the real novelty-curve figure (one line per source).

tually" occurs ∼6× more often in Claude than GPT-4o on HC3); this conversational lean is a secondary signal beneath the structural one.

### 5.6 Separability

An interpretable classifier confirms a stable, learnable voice (Table 6). Stripping markdown reduces AUC only modestly, so the classifier is not merely reading layout; and its top coefficients are exactly the features of §5.2 (burstiness, function-word density, em-dash, question rate), an independent-method confirmation. Raw length is not a feature, and residualizing every feature on response length barely changes separation (prose-only AUC 0.946→0.931 on HC3, 0.867→0.861 on AlpacaEval), so the result is not a length artifact.

### 5.7 Arbitrary-length spans and $n$-gram novelty

Suffix-automaton mining (Merrill et al., 2024) gives the most concrete form of the signature. Within each track, spans up to *eight words* occur dozens of times in Claude and *zero* times across that track's non-Claude sources (HC3: humans + 2 models; AlpacaEval: 6 models, no human): on HC3, *would you like me to explain any* (×20) and *would you like me to go deeper into* (×14); on AlpacaEval, *would you like me to expand on any* (×7). Claude also has the highest $n$-gram novelty of any source (Figure 2; e.g. bigram novelty 0.519 vs. 0.27–0.36 for the six AlpacaEval models): its exact phrasing overlaps least with the pool. We read novelty as relative distinctiveness, not absolute creativity (see Limitations).

### 5.8 Secondary analyses (in progress)

Three planned analyses are not yet included; designs and placeholder tables appear in Appendix D.

- **Effect of reasoning effort/thinking.** Whether the fingerprint intensifies or attenuates across `effort`∈ {low, medium, high} and with thinking disabled. *[To be completed; Appendix D.1.]*
- **Across Claude versions.** Intra-Claude drift across model versions (e.g. Opus/Sonnet/Haiku and older releases). *[To be completed; Appendix D.2.]*
- **Multi-turn and self-interaction.** Turn-dependent traits and the Claude-to-Claude "spiritual bliss" attractor reported in (Anthropic, 2025). *[To be completed; Appendix D.3.]*

## 6 Discussion

Across two corpora, five methods, and eight comparison sources, the same fingerprint recurs and survives every control: Claudespeak is markdown-scaffolded, rhythmically varied, content-dense, and engagement-and-offer-oriented prose—and specifically *not* the "delve" register. A methodological lesson follows: borrowed AI-word lists largely measure *GPT*, and characterizing a specific model requires data-driven discovery against topic-matched references.

## Limitations

We study a single Claude configuration (Opus 4.8, `effort=high`); the effort, version, and multi-turn axes are not yet measured (§5.8). Crucially, our coverage audit (§3.2) shows the corpus is single-turn, cooperative, and information-oriented, so the reported fingerprint is Claude's *single-turn informational* voice; context-gated behaviors (corrective concession, refusal, emotional support, code/creative registers) are out of scope and are the target of the planned expansion (English only). Reused contrast completions are their published vintages (e.g. gemini-pro 1.0, Qwen2-72B, GPT-4-turbo/4o 2024), modern-ish but not the absolute latest. $n$-gram novelty is measured relative to our corpus pool, not a large external reference, and may be inflated for Claude as the lone out-of-cluster (and only freshly generated) source. The HC3 human anchor is informal web text, so part of the human–Claude gap reflects register rather than authorship. The coverage audit's labels are produced by an LLM (the model under study), introducing a circularity; a human-validation pass on a labeled sample is pending. Finally, raw length is excluded from the classifier features and the feature effects are length-controlled (§5.3), but some prose features still correlate with length; we report a length-residualized classifier as a check (§5.6).

## Data and Reproducibility

We do not test mechanism, but the interactional, low-formality, offer-oriented profile is consistent with post-training (RLHF) that rewards helpful, engaging, and appropriately hedged responses. Code and the fully provenanced corpus (generations plus metadata, and the audit tooling) will be released at an anonymized URL. HC3 and AlpacaEval are public research artifacts used under their licenses; we redistribute only derived statistics and our own model generations.

## Ethics Statement

This work characterizes model writing style using public datasets and model outputs; it involves no human subjects or private data. Stylistic fingerprints can aid provenance and detection but could also inform evasion; we report aggregate, interpretable features rather than a deployable detector.

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## A Feature definitions

Full definitions of all interpretable features (rates, burstiness, markdown and rhetorical markers) are provided with the released code (`src/features.py`).

## B Full corpus statistics

All statistics here are reproducible via `src/dataset_stats.py` (volume, lengths, sentence mood, register) and `src/analyze_coverage.py` (intent).

| Axis | Value | HC3 | Alpaca |
|---|---|---|---|
| speech act | question | 150 | 106 |
| | directive | 46 | 93 |
| | assertion | 1 | 0 |
| register | neutral | 130 | 130 |
| | casual | 70 | 62 |
| | formal | 0 | 7 |
| tone | neutral | 165 | 175 |
| | emotional | 20 | 5 |
| | playful / urgent | 1/14 | 18/1 |

Table 7: Prompt variety along *pragmatic* speech act (LLM-labeled), register, and tone (HC3 $n$=200; AlpacaEval $n$=199). Assertions, formal register, and non-neutral tones are nearly absent. Topic and intent breakdowns: `coverage_labels.csv`. (Syntactic mood, which differs, is in §3.2.)

**Response length by source (words).** HC3: human 120/77 (mean/median), ChatGPT 176/178, GPT-4o 228/239, Claude 254/267. AlpacaEval: deepseek-67b 186/184, gemini-pro 222/234, Claude 254/230, Qwen2-72B 260/259, gpt-4-turbo 272/285, gpt-4o 280/285, Llama-3-70B 297/308.

**Volume by domain (prompts; total response words, all sources).** HC3: general 80/47k, finance 40/40k, cs_ai 40/35k, medicine 40/33k. AlpacaEval: selfinstruct 64/79k, oasst 49/96k, helpful_base 35/70k, koala 35/67k, vicuna 17/43k.

**Sentence mood (% of response sentences).** Interrogative sentences: Claude 17% (HC3) / 10% (AlpacaEval) vs. 1–4% for all other sources; declarative 75% (Claude, HC3) vs. 95–97% for others. Imperative and exclamatory are $\leq$2% and $\leq$7% respectively.

**Register.** First-/second-person pronoun rates per 100 words and Flesch Reading Ease are in `stats_register.csv`; Claude shows the lowest Flesch ( $\approx$40, denser/harder) of all sources on both tracks.

**Prompt variety (audit, counts of 200 per track).** The prompts are narrow on every axis (Table 7): almost all are *questions* or *directives* (assertions $\leq$1); tone is overwhelmingly *neutral* (82–87%); register is neutral/casual with almost no *formal* prompts; and topics concentrate (HC3: health 28%, science 24%, finance 20%). This uniformity—not insufficient sample size—is the coverage problem the expansion targets.

| Track | Domain | burst. | func↓ | em-d. | ques. |
|---|---|---|---|---|---|
| HC3 | cs/ai | 2.50 | −1.86 | 1.12 | 1.02 |
| HC3 | finance | 2.11 | −1.75 | 0.94 | 1.55 |
| HC3 | general | 1.46 | −0.97 | 0.95 | 0.17 |
| HC3 | medicine | 2.65 | −1.28 | 2.41 | 0.72 |
| Alpaca | helpful | 1.19 | −0.81 | 0.66 | 0.85 |
| Alpaca | koala | 0.99 | −0.43 | 2.02 | 0.25 |
| Alpaca | oasst | 1.46 | −0.77 | 1.38 | 1.26 |
| Alpaca | selfinst. | 0.56 | −0.40 | 0.26 | 0.29 |
| Alpaca | vicuna | 1.98 | −1.17 | 3.22 | 2.21 |

Table 8: Per-domain Claude-vs-others Cohen's $d$, prose only. burst. = sentence burstiness; func↓ = function-word rate (negative = Claude denser); em-d. = em-dash; ques. = question rate. Uniform signs across domains.

**Coverage expansion (planned).** To move beyond the single-turn informational slice (§3.2) we will (i) sample real first-turn prompts from WildChat / LMSYS-Chat-1M to populate under-covered intents (code, creative, opinion, emotional); (ii) *construct* multi-turn corrective scenarios (user challenges or corrects Claude) to elicit concession / push-back behavior; and (iii) add a small, defensively-framed sensitive/refusal set. Targets are ~100–150 prompts per intent cell and ~80–120 corrective scenarios; the audit tooling reruns on the expanded corpus.

## C Per-domain robustness

Claude-vs-pooled-others Cohen's $d$ (markdown-stripped prose) computed *within* each domain (Table 8). The signs are uniform across all nine domains—burstiness and em-dash/question positive, function-word rate negative (Claude denser)—so the fingerprint is not driven by any single topic.

## D Placeholders for secondary analyses

### D.1 Effect of reasoning effort

*Planned design.* Regenerate the 200 HC3 prompts (and 200 AlpacaEval prompts) under `effort`$\in$ {low, medium, high} and with thinking disabled, holding all else fixed; recompute the §5.2 features and the §5.6 classifier within each setting; report whether burstiness, density, em-dash, and the offer-closer scale with effort. *[Results table to be inserted.]*

### D.2 Across Claude versions

*Planned design.* Generate the same prompts with additional Claude versions (e.g. Sonnet/Haiku, and an older 3.x), then measure intra-Claude drift in the fingerprint features and the offer-closer rate. *[Results table to be inserted.]*

### D.3 Multi-turn and self-interaction

*Planned design.* (i) Two-turn parallel probes (e.g. MT-Bench-style) to measure turn-dependent traits comparatively; (ii) elicited Claude-to-Claude conversations to test the "spiritual bliss" attractor (Anthropic, 2025) quantitatively (e.g. trajectory of consciousness/gratitude vocabulary and vocabulary collapse over turns). *[Results table to be inserted.]*